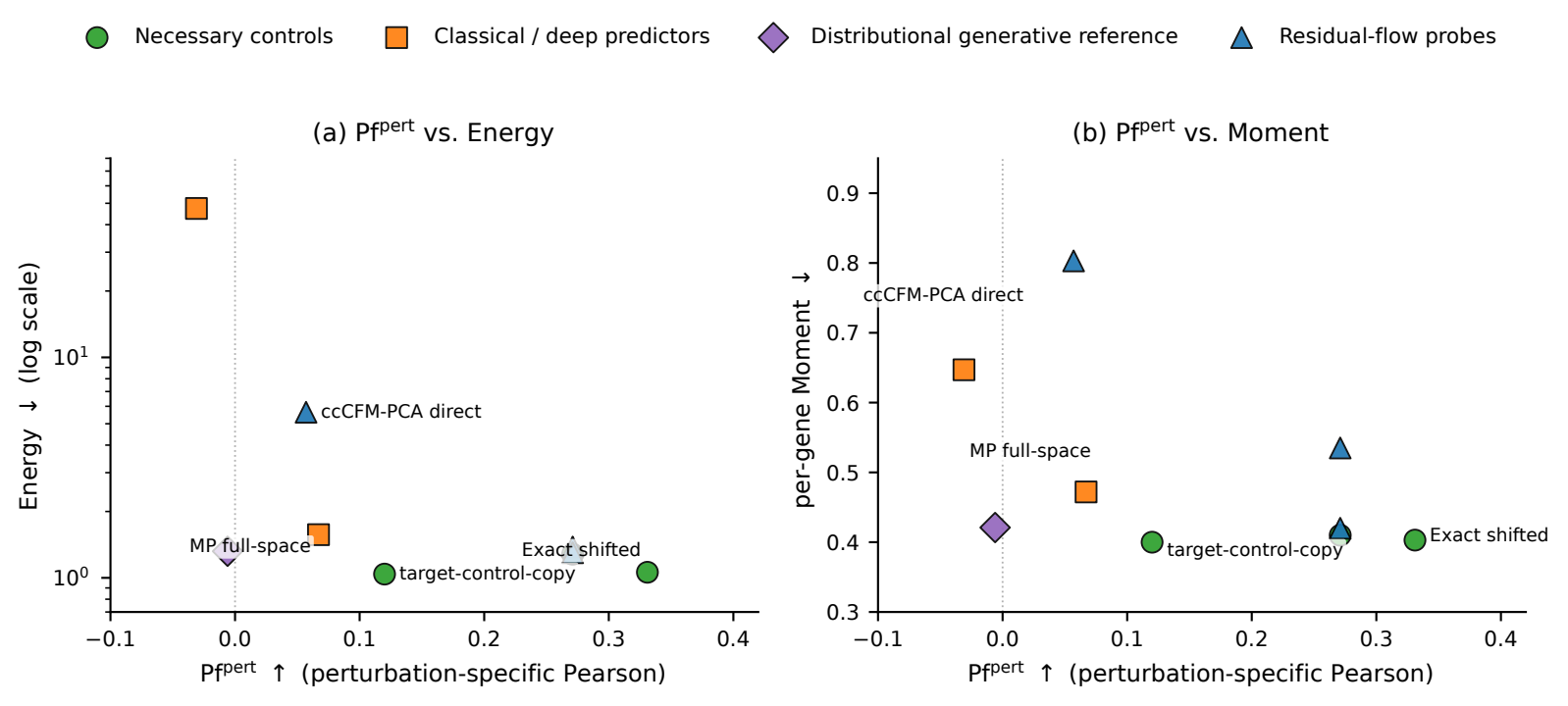
Necessary controls
Classical / deep predictors
Distributional generative reference
Residual-flow probes
(a) Pf^pert vs. Energy
Energy ↓ (log scale)
10^1
10^0
ccCFM-PCA direct
MP full-space
Exact shifted
target-control-copy
−0.1
0.0
0.1
0.2
0.3
0.4
Pf^pert ↑ (perturbation-specific Pearson)
(b) Pf^pert vs. Moment
per-gene Moment ↓
0.9
0.8
0.7
0.6
0.5
0.4
0.3
ccCFM-PCA direct
MP full-space
target-control-copy
Exact shifted
−0.1
0.0
0.1
0.2
0.3
0.4
Pf^pert ↑ (perturbation-specific Pearson)